

Table 1: Number of models with at least one structural or logical defect (# of S/L) and mean specification-alignment precision (P), recall (R), and F_1 before refinement and after one round per diagnostic approach. Bold marks the highest P, R, and F_1 within each dataset–LLM setting. The values averaged over more than one run are provided with their standard deviation values.

Approach	Industry								Paged							
	Gemini				Qwen				Gemini				Qwen			
	# S/L	P	R	F_1	# S/L	P	R	F_1	# S/L	P	R	F_1	# S/L	P	R	F_1
No refinement	157 $\pm$ 0.0	0.32 $\pm$ 0.004	0.44 $\pm$ 0.001	0.36 $\pm$ 0.004	157 $\pm$ 0.0	0.29 $\pm$ 0.003	0.46 $\pm$ 0.003	0.32 $\pm$ 0.001	203 $\pm$ 0.0	0.41 $\pm$ 0.003	0.47 $\pm$ 0.003	0.43 $\pm$ 0.002	203 $\pm$ 0.0	0.38 $\pm$ 0.002	0.47 $\pm$ 0.003	0.41 $\pm$ 0.002
MODEL-INTERNAL	4	0.59	0.80	0.63	72 $\pm$ 3.5	0.40 $\pm$ 0.013	0.67 $\pm$ 0.022	0.46 $\pm$ 0.015	15	0.72	0.83	0.74	51 $\pm$ 2.5	0.61 $\pm$ 0.009	0.79 $\pm$ 0.011	0.66 $\pm$ 0.008
ELEMENT ALIGNMENT	4	0.59	0.73	0.61	69 $\pm$ 8.5	0.43 $\pm$ 0.021	0.66 $\pm$ 0.029	0.48 $\pm$ 0.022	3	0.75	0.85	0.77	62 $\pm$ 3.4	0.64 $\pm$ 0.004	0.77 $\pm$ 0.011	0.67 $\pm$ 0.006
DIRECT LLM	5	0.55	0.72	0.58	86 $\pm$ 4.1	0.41 $\pm$ 0.013	0.58 $\pm$ 0.012	0.44 $\pm$ 0.009	5	0.78	0.82	0.77	84 $\pm$ 5.3	0.64 $\pm$ 0.007	0.71 $\pm$ 0.011	0.65 $\pm$ 0.008
IntentTrace	19	0.71	0.82	0.72	44 $\pm$ 4.0	0.55 $\pm$ 0.010	0.81 $\pm$ 0.012	0.61 $\pm$ 0.009	15	0.86	0.85	0.83	51 $\pm$ 3.7	0.72 $\pm$ 0.011	0.81 $\pm$ 0.006	0.74 $\pm$ 0.006

Table 2: Pairwise statistical comparison after one refinement round per diagnostic approach for precision, recall and F1 under IntentTrace, based on the results in Table 1. Each round is compared against the one immediately before it. (S), (M) and (L) denote small, medium and large effect size. Yellow cells highlight comparisons where the newer round shows a statistically significant improvement over the previous round.

Dataset	LLM	Metric	Model-Internal vs No Refinement		Element Alignment vs No Refinement		Direct LLM vs No Refinement		IntentTrace vs Model-Internal		IntentTrace vs Element Alignment		IntentTrace vs Direct LLM	
			p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$
Industry	Gemini	Precision	1.38E-25	0.71 (M)	3.95E-20	0.71 (L)	1.02E-14	0.68 (M)	1.09E-11	0.65 (M)	1.83E-11	0.64 (M)	3.86E-15	0.69 (M)
		Recall	1.38E-25	0.66 (M)	1.61E-16	0.60 (S)	1.22E-14	0.60 (S)	6.78E-03	0.60 (S)	1.55E-11	0.71 (M)	2.93E-12	0.70 (M)
		F_1	1.38E-25	0.69 (M)	8.11E-18	0.67 (M)	6.23E-14	0.65 (M)	1.15E-09	0.62 (S)	4.85E-12	0.65 (M)	1.58E-16	0.68 (M)
	Qwen	Precision	4.91E-64	0.61 (S)	2.10E-57	0.62 (S)	2.00E-33	0.60 (S)	4.25E-65	0.62 (S)	5.62E-48	0.60 (S)	3.19E-43	0.62 (S)
		Recall	4.91E-64	0.58 (S)	1.84E-33	0.56 (S)	1.75E-08	0.51	5.01E-44	0.64 (M)	2.56E-60	0.69 (M)	3.28E-94	0.75 (L)
		F_1	4.91E-64	0.61 (S)	4.59E-47	0.61 (S)	1.79E-21	0.58 (S)	2.52E-66	0.62 (S)	2.10E-57	0.61 (S)	8.31E-63	0.64 (M)
PAGED	Gemini	Precision	2.57E-31	0.67 (M)	3.25E-37	0.69 (M)	2.58E-36	0.71 (M)	1.55E-19	0.67 (M)	5.02E-16	0.64 (M)	8.34E-10	0.61 (S)
		Recall	2.57E-31	0.66 (M)	9.11E-36	0.67 (M)	1.81E-31	0.64 (S)	0.17	0.54	0.71	0.53	1.88E-04	0.59 (S)
		F_1	2.57E-31	0.66 (M)	2.39E-37	0.68 (M)	1.20E-34	0.68 (M)	1.19E-13	0.63 (S)	4.96E-09	0.60 (S)	8.52E-10	0.60 (S)
	Qwen	Precision	9.01E-121	0.65 (M)	1.56E-137	0.66 (M)	1.51E-120	0.66 (M)	1.08E-47	0.61 (S)	2.79E-32	0.58 (S)	4.96E-19	0.55
		Recall	5.28E-122	0.64 (M)	2.66E-105	0.63 (S)	5.56E-73	0.60 (S)	1.31E-04	0.54	9.04E-10	0.55	8.69E-29	0.58 (S)
		F_1	9.01E-121	0.64 (M)	3.15E-136	0.65 (M)	7.85E-109	0.64 (M)	8.45E-41	0.59 (S)	1.40E-29	0.56 (S)	3.04E-26	0.56

Table 3: Mean precision (P), recall (R), and F_1 after one refinement round per diagnostic approach under the ground-truth-based measures of LADEX and AbsCon. Structurally or logically unsound models score zero, as in Table 1. Bold marks the highest P, R, and F_1 within each measure–dataset–LLM setting. The values averaged over more than one run are provided with their standard deviation values.

Scorer	Approach	Industry						Paged					
		Gemini			Qwen			Gemini			Qwen		
		P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1
AbsCon	No refinement	0.27 $\pm$ 0.000	0.40 $\pm$ 0.000	0.32 $\pm$ 0.000	0.27 $\pm$ 0.000	0.40 $\pm$ 0.000	0.32 $\pm$ 0.000	0.35 $\pm$ 0.000	0.36 $\pm$ 0.000	0.35 $\pm$ 0.000	0.35 $\pm$ 0.000	0.36 $\pm$ 0.000	0.35 $\pm$ 0.000
	MODEL-INTERNAL	0.57	0.84	0.67	0.44 $\pm$ 0.006	0.64 $\pm$ 0.010	0.52 $\pm$ 0.008	0.67	0.65	0.65	0.61 $\pm$ 0.004	0.59 $\pm$ 0.004	0.59 $\pm$ 0.004
	ELEMENT ALIGNMENT	0.59	0.84	0.68	0.45 $\pm$ 0.018	0.64 $\pm$ 0.024	0.53 $\pm$ 0.020	0.70	0.67	0.68	0.60 $\pm$ 0.005	0.57 $\pm$ 0.007	0.58 $\pm$ 0.006
	DIRECT LLM	0.59	0.83	0.68	0.43 $\pm$ 0.010	0.58 $\pm$ 0.012	0.49 $\pm$ 0.010	0.76	0.60	0.66	0.61 $\pm$ 0.009	0.49 $\pm$ 0.011	0.53 $\pm$ 0.010
	IntentTrace	0.56	0.79	0.65	0.51 $\pm$ 0.008	0.71 $\pm$ 0.012	0.59 $\pm$ 0.010	0.68	0.65	0.65	0.63 $\pm$ 0.007	0.58 $\pm$ 0.007	0.59 $\pm$ 0.007
LADEX (B-MATCH)	No refinement	0.43 $\pm$ 0.000	0.41 $\pm$ 0.000	0.41 $\pm$ 0.000	0.43 $\pm$ 0.000	0.41 $\pm$ 0.000	0.41 $\pm$ 0.000	0.42 $\pm$ 0.000	0.45 $\pm$ 0.000	0.42 $\pm$ 0.000	0.42 $\pm$ 0.000	0.45 $\pm$ 0.000	0.42 $\pm$ 0.000
	MODEL-INTERNAL	0.86	0.86	0.84	0.67 $\pm$ 0.009	0.66 $\pm$ 0.013	0.65 $\pm$ 0.011	0.81	0.88	0.80	0.74 $\pm$ 0.006	0.80 $\pm$ 0.005	0.73 $\pm$ 0.006
	ELEMENT ALIGNMENT	0.90	0.78	0.80	0.70 $\pm$ 0.027	0.63 $\pm$ 0.024	0.64 $\pm$ 0.026	0.86	0.91	0.85	0.74 $\pm$ 0.007	0.78 $\pm$ 0.009	0.73 $\pm$ 0.007
	DIRECT LLM	0.86	0.83	0.82	0.65 $\pm$ 0.010	0.56 $\pm$ 0.013	0.58 $\pm$ 0.011	0.88	0.87	0.85	0.73 $\pm$ 0.013	0.72 $\pm$ 0.012	0.71 $\pm$ 0.014
	IntentTrace	0.82	0.70	0.72	0.78 $\pm$ 0.014	0.68 $\pm$ 0.009	0.70 $\pm$ 0.010	0.88	0.88	0.85	0.80 $\pm$ 0.014	0.80 $\pm$ 0.007	0.77 $\pm$ 0.010

Table 4: Pairwise statistical comparison after one refinement round per diagnostic approach for precision, recall and F1 under the ground-truth-based measure of AbsCon, based on the results in Table 3. Each round is compared against the one immediately before it. (S), (M) and (L) denote small, medium and large effect size. Yellow cells highlight comparisons where the newer round shows a statistically significant improvement over the previous round.

Dataset	LLM	Metric	Model-Internal vs No Refinement		Element Alignment vs No Refinement		Direct LLM vs No Refinement		IntentTrace vs Model-Internal		IntentTrace vs Element Alignment		IntentTrace vs Direct LLM	
			p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$
Industry	Gemini	Precision	1.90E-26	0.78 (L)	8.03E-41	0.80 (L)	5.79E-40	0.80 (L)	5.14E-06	0.52	8.91E-03	0.47	0.07	0.47
		Recall	1.90E-26	0.77 (L)	2.94E-26	0.76 (L)	5.86E-25	0.75 (L)	4.17E-05	0.45	0.45	0.47	0.44	0.48
		F_1	1.90E-26	0.78 (L)	7.47E-37	0.80 (L)	2.20E-34	0.80 (L)	0.01	0.51	0.07	0.47	0.15	0.47
	Qwen	Precision	9.80E-71	0.66 (M)	2.15E-103	0.67 (M)	6.22E-76	0.66 (M)	3.91E-61	0.58 (S)	1.01E-13	0.55	8.34E-07	0.56
		Recall	9.80E-71	0.64 (M)	2.34E-45	0.64 (S)	1.58E-17	0.59 (S)	0.17	0.52	7.05E-04	0.54	5.85E-44	0.61 (S)
		F_1	9.80E-71	0.66 (M)	1.16E-81	0.67 (M)	1.04E-45	0.64 (S)	1.23E-26	0.56 (S)	3.95E-08	0.55	2.57E-22	0.58 (S)
PAGED	Gemini	Precision	3.43E-32	0.71 (L)	1.31E-41	0.74 (L)	8.52E-60	0.79 (L)	2.37E-05	0.53	0.39	0.50	9.38E-30	0.38 (S)
		Recall	3.43E-32	0.70 (M)	1.30E-38	0.71 (M)	1.09E-20	0.65 (M)	0.01	0.49	2.68E-04	0.48	2.11E-18	0.60 (S)
		F_1	3.43E-32	0.70 (M)	3.76E-40	0.72 (L)	9.28E-31	0.71 (M)	0.48	0.51	0.06	0.48	0.30	0.50
	Qwen	Precision	1.15E-125	0.67 (M)	1.23E-137	0.67 (M)	2.40E-168	0.69 (M)	9.05E-16	0.53	1.17E-09	0.52	1.28E-16	0.47
		Recall	1.15E-125	0.66 (M)	5.64E-107	0.64 (S)	3.03E-27	0.58 (S)	5.91E-12	0.48	0.06	0.50	8.54E-77	0.60 (S)
		F_1	1.15E-125	0.66 (M)	7.35E-118	0.65 (M)	1.12E-55	0.62 (S)	0.47	0.50	0.12	0.51	1.58E-16	0.55

Table 5: Pairwise statistical comparison after one refinement round per diagnostic approach for precision, recall and F1 under the ground-truth-based measure of LADEX, based on the results in Table 3 . Each round is compared against the one immediately before it. (S), (M) and (L) denote small, medium and large effect size. Yellow cells highlight comparisons where the newer round shows a statistically significant improvement over the previous round.

Dataset	LLM	Metric	Model-Internal vs No Refinement		Element Alignment vs No Refinement		Direct LLM vs No Refinement		Intent/Trace vs Model-Internal		Intent/Trace vs Element Alignment		Intent/Trace vs Direct LLM	
			p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$
Industry	Gemini	Precision	1.56E-27	0.74 (L)	1.23E-31	0.76 (L)	3.53E-29	0.74 (L)	0.99	0.50	4.38E-04	0.46	0.67	0.49
		Recall	1.22E-27	0.77 (L)	1.21E-24	0.72 (L)	1.22E-27	0.75 (L)	3.96E-14	0.37 (S)	4.38E-04	0.45	6.47E-10	0.40 (S)
		F1	9.70E-27	0.75 (L)	8.00E-27	0.73 (L)	1.56E-27	0.74 (L)	8.09E-07	0.40 (S)	2.33E-03	0.45	2.67E-05	0.42 (S)
	Qwen	Precision	2.47E-73	0.63 (S)	4.41E-72	0.65 (M)	7.66E-48	0.63 (S)	2.07E-23	0.58 (S)	8.08E-13	0.55	1.65E-20	0.57 (S)
		Recall	1.35E-76	0.65 (M)	3.07E-56	0.63 (S)	1.82E-27	0.59 (S)	0.35	0.49	1.14E-05	0.53	3.45E-20	0.58 (S)
		F1	2.20E-71	0.64 (M)	4.11E-60	0.63 (S)	7.56E-31	0.59 (S)	1.89E-03	0.51	1.33E-07	0.54	5.91E-21	0.58 (S)
PAGED	Gemini	Precision	1.73E-33	0.72 (L)	3.59E-38	0.75 (L)	2.74E-41	0.76 (L)	9.70E-07	0.57 (S)	0.04	0.53	0.61	0.50
		Recall	2.57E-35	0.73 (L)	3.59E-38	0.74 (L)	1.73E-36	0.72 (L)	0.85	0.51	0.01	0.49	0.40	0.53
		F1	1.31E-32	0.72 (L)	4.04E-38	0.74 (L)	1.74E-39	0.74 (L)	3.57E-05	0.57 (S)	0.40	0.53	0.61	0.52
	Qwen	Precision	3.32E-130	0.68 (M)	1.27E-128	0.68 (M)	1.52E-116	0.67 (M)	1.77E-11	0.55	3.31E-10	0.54	2.55E-09	0.52
		Recall	4.94E-141	0.69 (M)	5.48E-116	0.67 (M)	5.59E-88	0.64 (S)	0.42	0.49	0.06	0.50	5.61E-14	0.54
		F1	2.65E-127	0.68 (M)	9.15E-124	0.68 (M)	5.42E-110	0.66 (M)	3.10E-07	0.54	1.05E-06	0.53	5.52E-10	0.53

Table 6: Number of models with at least one structural or logical defect (# of S/L) and mean specification-alignment precision (P), recall (R), and F_1 before refinement and after one round per diagnostic approach. Each specification is collapsed to a single score per approach. Bold marks the highest P , R , and F_1 within each dataset–LLM setting; multi-run averages include standard deviations.

Approach	Industry								Paged							
	Gemini				Qwen				Gemini				Qwen			
	S/L	P	R	F1	S/L	P	R	F1	S/L	P	R	F1	S/L	P	R	F1
No refinement	10.65 ± 0.00	0.32 ± 0.004	0.43 ± 0.002	0.35 ± 0.004	10.65 ± 0.00	0.29 ± 0.003	0.45 ± 0.003	0.32 ± 0.001	86.73 ± 0.00	0.43 ± 0.004	0.48 ± 0.003	0.44 ± 0.003	86.73 ± 0.00	0.39 ± 0.002	0.48 ± 0.003	0.42 ± 0.002
MODEL-INTERNAL	0.33	0.57	0.80	0.62	5.12 ± 0.27	0.40 ± 0.013	0.65 ± 0.021	0.45 ± 0.014	7.70	0.72	0.83	0.74	22.18 ± 1.54	0.62 ± 0.010	0.79 ± 0.012	0.66 ± 0.009
ELEMENT ALIGNMENT	0.36	0.58	0.73	0.60	4.79 ± 0.53	0.43 ± 0.021	0.65 ± 0.028	0.47 ± 0.022	0.92	0.76	0.86	0.77	27.00 ± 2.24	0.64 ± 0.006	0.77 ± 0.015	0.67 ± 0.008
DIRECT LLM	0.38	0.54	0.73	0.57	5.86 ± 0.26	0.41 ± 0.011	0.57 ± 0.015	0.44 ± 0.008	2.25	0.79	0.82	0.78	37.28 ± 2.86	0.64 ± 0.012	0.71 ± 0.011	0.65 ± 0.011
Intent/Trace	1.47	0.70	0.81	0.70	3.17 ± 0.31	0.55 ± 0.013	0.80 ± 0.015	0.60 ± 0.013	6.45	0.86	0.85	0.84	21.87 ± 2.60	0.73 ± 0.010	0.81 ± 0.012	0.74 ± 0.007

Table 7: Pairwise statistical comparison of consecutive refinement rounds per diagnostic approach for precision, recall, and F_1 under IntentTrace based on the results in Table 6. (S), (M), and (L) denote small, medium, and large effect sizes; yellow cells highlight statistically significant improvements in the newer round over the previous.

Dataset	LLM	Metric	Model-Internal vs No Refinement		Element Alignment vs No Refinement		Direct LLM vs No Refinement		IntentTrace vs Model-Internal		IntentTrace vs Element Alignment		IntentTrace vs Direct LLM	
			p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$	p	$\hat{A}_{12}$
Industry	Gemini	Precision	1.37E-05	0.78 (L)	1.37E-05	0.81 (L)	5.79E-04	0.76 (L)	3.43E-05	0.69 (M)	1.37E-05	0.69 (M)	1.72E-05	0.76 (L)
		Recall	1.37E-05	0.89 (L)	1.34E-04	0.84 (L)	3.92E-04	0.83 (L)	0.67	0.58 (S)	0.02	0.74 (L)	0.01	0.74 (L)
		F1	1.37E-05	0.79 (L)	2.45E-05	0.79 (L)	5.47E-04	0.76 (L)	5.79E-04	0.67 (M)	3.43E-05	0.69 (M)	3.00E-05	0.74 (L)
	Qwen	Precision	1.58E-04	0.64 (S)	4.90E-06	0.68 (M)	3.63E-04	0.67 (M)	1.14E-05	0.66 (M)	4.90E-06	0.64 (M)	4.90E-06	0.67 (M)
		Recall	1.58E-04	0.72 (L)	3.12E-05	0.72 (L)	0.02	0.64 (M)	5.44E-05	0.75 (L)	4.90E-06	0.76 (L)	4.90E-06	0.90 (L)
		F1	1.58E-04	0.65 (M)	1.14E-05	0.67 (M)	1.52E-03	0.67 (M)	1.72E-05	0.69 (M)	4.90E-06	0.66 (M)	4.90E-06	0.72 (L)
PAGED	Gemini	Precision	3.17E-21	0.71 (M)	5.01E-22	0.73 (L)	5.01E-22	0.75 (L)	1.27E-12	0.67 (M)	3.70E-10	0.65 (M)	1.54E-07	0.61 (S)
		Recall	6.36E-21	0.72 (L)	3.17E-21	0.73 (L)	1.37E-20	0.71 (M)	0.05	0.55	0.99	0.54	1.22E-03	0.59 (S)
		F1	2.35E-21	0.71 (M)	5.01E-22	0.73 (L)	5.01E-22	0.73 (L)	1.05E-09	0.63 (S)	8.09E-06	0.60 (S)	5.41E-08	0.60 (S)
	Qwen	Precision	1.80E-19	0.68 (M)	3.36E-22	0.70 (M)	2.32E-20	0.70 (M)	9.28E-13	0.62 (S)	4.02E-12	0.58 (S)	7.29E-10	0.56 (S)
		Recall	1.80E-19	0.68 (M)	1.13E-18	0.67 (M)	3.81E-14	0.63 (S)	0.09	0.51	2.04E-04	0.54	9.15E-12	0.61 (S)
		F1	1.80E-19	0.68 (M)	1.04E-21	0.69 (M)	3.33E-19	0.68 (M)	3.42E-10	0.59 (S)	8.68E-10	0.57 (S)	3.58E-11	0.58 (S)